Jongseo Lee

Researcher, Vision and Learning Lab, Kyung Hee University, South Korea

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RESEARCH INTERESTS

- **Trustworthy video understanding:** interpretable concept-based models; diagnosing and improving Video-LLMs (motion and physical reasoning)
- **Video world models for Physical AI:** learning to perceive and predict physical dynamics from video (intuitive physics)
- **Video representation learning:** multimodal (audio-visual, video-language) and continual learning

EDUCATION

M.S. in Computer Science and Engineering, Kyung Hee University — Sep. 2023 – Aug. 2025
Advisor: Prof. Jinwoo Choi · GPA 4.21 / 4.3

B.S. in Biomedical Engineering & Electronic Engineering, Kyung Hee University — Mar. 2017 – Aug. 2023
Double major · GPA 3.95 / 4.3

PUBLICATIONS

* *equal contribution*

Conferences & Journals

1. **Jongseo Lee***, Hyuntak Lee*, Sunghun Kim, Sooa Kim, Jihoon Chung, Jinwoo Choi, Which Way Did It Move? Diagnosing and Overcoming Directional Motion Blindness in Video-LLMs. Accepted to *Neural Information Processing Systems* (**NeurIPS**) 2026. Also at the CompLearn Workshop, ICML 2026 (non-archival). paper project code
2. Wooil Lee*, Junpyo Hong*, **Jongseo Lee**, Tuomas Oikarinen, Tsui-Wei Weng, Jinwoo Choi, TRACE: Trajectory-Aware Concepts for Explainable Video Understanding. Accepted to *Neural Information Processing Systems* (**NeurIPS**) 2026.
3. **Jongseo Lee**, Wooil Lee, Gyeong-Moon Park, Seong Tae Kim, Jinwoo Choi, Disentangled Concepts Speak Louder Than Words: Explainable Video Action Recognition. *Neural Information Processing Systems* (**NeurIPS**) 2025, *Spotlight* (top 3.5%). paper code
4. **Jongseo Lee***, Kyungho Bae*, Kyle Min, Gyeong-Moon Park, Jinwoo Choi, ESSENTIAL: Episodic and Semantic Memory Integration for Video Class-Incremental Learning. *IEEE/CVF International Conference on Computer Vision* (**ICCV**) 2025, *Highlight* (top 2.9%). paper code
5. **Jongseo Lee***, Joohyun Chang*, Dongho Lee, Jinwoo Choi, CA^{2}ST: Cross-Attention in Audio, Space, and Time for Holistic Video Recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (**TPAMI**) 2026. paper
6. **Jongseo Lee**, Wooil Lee, Gyeong-Moon Park, Seong Tae Kim, Jinwoo Choi, PCBEAR: Pose Concept Bottleneck for Explainable Action Recognition. *CVPR Workshop on Explainable AI for Computer Vision* (**XAI4CV**) 2025, *Spotlight* (top 16.7%). paper
7. **Jongseo Lee***, Dongho Lee*, Jinwoo Choi, CAST: Cross-Attention in Space and Time for Video Action Recognition. *Neural Information Processing Systems* (**NeurIPS**) 2023. paper poster code

Preprints

8. **Jongseo Lee***, Geo Ahn*, Seong Tae Kim, Jinwoo Choi, PCEvE: Part Contribution Evaluation Based Model Explanation for Human Figure Drawing Assessment and Beyond. *arXiv preprint*, 2024. *Excellent Paper Award*, KCC 2024 Workshop on Explainable AI. paper

Non-Archival Workshop Presentations

- **Jongseo Lee***, Joohyun Chang*, Dongho Lee, Jinwoo Choi, BiAdA: Holistic Video Recognition via Bidirectional Adapter Fusion. *ICCV Workshop on Long Video Foundations* (**LongVidFoundations**) 2025, poster.

Domestic Publications

- **Jongseo Lee**, Wooil Lee, Jinwoo Choi, Disentangled Concepts for Video Action Recognition. *Workshop on Image Processing and Image Understanding* (**IPIU**) 2026, *Excellent Paper Award*.
- Soohyun Park, **Jongseo Lee**, Jinwoo Choi, Efficient Video Class-Incremental Learning via Class-Token Retraining (in Korean). *KIISE Korea Computer Congress (KCC) 2024, Undergraduate Paper Competition*, *Excellence Award* (mentored undergraduate first author).
- Joohyun Chang, **Jongseo Lee**, Jinwoo Choi, Audio-Video Cross-Attention for Effective Video Action Recognition (in Korean). *KCC 2024, Undergraduate Paper Competition*, *Encouragement Award* (mentored undergraduate first author).

– **Jongseo Lee**, Su Hyeon Kim, Sun Woong Jang, Jun Yeong Moon, Doug Young Suh, Metaverse Interface with Haptic and Rigid Sense Feedback at a Low Cost. *Journal of Appropriate Technology*, 8(2), 2022. paper

RESEARCH EXPERIENCE

Researcher (Post-M.S.), Vision and Learning Lab, Kyung Hee University Sep. 2025 – Present

- **Video-LLM motion diagnosis** (Which Way Did It Move?): showed that 9 Video-LLMs answer direction questions near chance though a linear probe reads direction at 95.3%; proposed DeltaDirect (27.6% → 85.9%, no inference overhead).
- **Sparse autoencoders for video models** (with Princeton, Prof. Olga Russakovsky; ongoing): finding bias features.
- **Intuitive physics in video world models** (ongoing): can they learn it from observation alone?
- **Trajectory concepts** (TRACE, with UC San Diego, Prof. Tsui-Wei Weng): from pose to optical-flow concepts.
- **Industry project** (MetaRocket, 2026): skeleton-based recognition of multi-person K-POP choreography.

Graduate Research Assistant, Vision and Learning Lab, Kyung Hee University Sep. 2023 – Aug. 2025

- **Explainable action recognition** (DANCE, PCBEAR): separated motion, object, and scene concepts in a concept bottleneck; users preferred its explanations in over 70% of comparisons.
- **Video class-incremental learning** (ESSENTIAL, with Kyle Min, Intel Labs): stored sparse features and reconstructed dense ones, beating JPEG-frame replay with about 27× less memory.
- **ETRI-funded projects** (2023–2025): explainable ASD screening from drawings (PCEvE); reaction analysis in video.

Undergraduate Research Intern, Vision and Learning Lab, Kyung Hee University Feb. 2022 – Jul. 2023

- **Spatio-temporal expert fusion** (CAST): connected spatial (CLIP) and temporal (VideoMAE) experts through bottleneck cross-attention; later added an audio expert (CA^2ST).
- **National dataset project** (AI Hub, with GAZZILABS, 2022): recognition algorithms for group activity video.

HONORS & SCHOLARSHIPS

Top Reviewer, NeurIPS 2026 2026

Excellent Paper Award, IPIU 2026 (Disentangled Concepts for Video Action Recognition) 2026

Spotlight / Highlight papers: NeurIPS 2025 Spotlight (top 3.5%) · ICCV 2025 Highlight (top 2.9%) · CVPR 2025 Workshop XAI4CV Spotlight (top 16.7%) 2025

Excellent Paper Award, KCC 2024 Workshop on Explainable AI (PCEvE) 2024

Kyung Hee Scholarship for Outstanding Undergraduate Alumni Sep. 2023 – Aug. 2025
Full M.S. tuition for 4 semesters (approx. $21,600 total)

Woowa Brothers Scholarship, JUMP Foundation (approx. $2,900) 2022

ACADEMIC SERVICE

Reviewer NeurIPS 2026 (*Top Reviewer*) · AAAI 2025 · CVPR 2026 Workshop (XAI4CV) · ICML 2026 Workshop (CompLearn)

SKILLS & MILITARY SERVICE

Programming Python, C/C++, Bash · PyTorch · Linux, LaTeX · large-scale video/multimodal training and evaluation

Military Republic of Korea Army, Sergeant (active duty), Jul. 2018 – Mar. 2020; honorably discharged

REFEREES

Jinwoo Choi
Assistant Professor, Kyung Hee University
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M.S. advisor, current supervisor

Olga Russakovsky
Associate Professor, Princeton University
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Collaborator (video model interpretability)

Tsui-Wei (Lily) Weng
Assistant Professor, UC San Diego
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Collaborator (TRACE)

Kyle Min
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